

Name	Dr. Aman Kumari Dahiya
Designation	Associate Professor & Associate Dean Infrastructure, MSIT
Qualifications	Ph.D., M.Tech, M.B.A., B.E.
Email	amandahiya@msit.in
Area of Interest	Artificial Intelligence, IOT based applications, Design of Microwave & Millimetre-wave waveguides, Antenna Designing,
Experience	20 years



Awards and Recognitions:

1. Received **the Best Paper Award** at the National Conference on Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), organized by MSIT on 18–19 November 2025.
2. **Received Research Excellence Award** for valuable contribution in publishing Science citation Indexed (SCIE) research paper on Annual Day 2023 at MSIT.
3. **Received Research Excellence Award** for valuable contribution in completing IEEE Research Project by Maharaja Surajmal Institute of Technology on annual day in January 2020.
4. Awarded “**certificate of Excellence**” for best lecture presentation in National level seminar held on 24th March, 2006 at N.C. College of Engineering, Israna, Panipat.
5. **College Topper** during B.E. Degree Programme.
6. Got **distinction** in **B.E** and **M.Tech**.
7. **School Topper** in Matriculation.

Key Contributions:

- Served as **Faculty Representative (2020-2022)** in **Board of Governors** of Maharaja Surajmal Institute of Technology.
- Successfully organized 7th National conference on Advances in Metrology **AdMet-2021** on March 5th-6th, 2021 successfully in association with CSIR-National Physical Laboratory (CSIR-NPL), Metrology Society of India (MSI) and National Accreditation Board for Testing & Calibration Laboratories (NABL) India in the capacity of the **Co-convener**
- Hosted a special issue on “Advances in Sensors and Measurements for Metrological Applications” in capacity of **guest editor of MAPAN (SCIE Indexed)**, July 2021.
- Edited springer book series on “Lecture Notes in Electrical Engineering” for **AdMet-2021** proceedings (**SCOPUS Indexed**), 2021.

- **Coordinator** of short term course on “Modelling & simulation using MATLAB” conducted by NITTTR, Chandigarh from 21st -25th March 2018 . [one week]
- Organized ENVA-2020 (Environmental Fest of MSIT) as **Co-convener**.
- **Chaired special session** in an International symposium on 11th -12th Nov. 2021 organized by Sikkim Manipal University.
- Effectively coordinated **stage arrangements** for major departmental and Institute-level events, including anchoring, table and floral arrangements, reception, and certificate distribution.
- Organized annual **Tech-Fest AVENSIS-2025** in the capacity of **Convener** successfully.
- Served as **Convener of the Media Cell** of the institute from 2015 to 2021 and was the Founder of the Broadcast Cell MSIT. Initiated and led various initiatives to establish and strengthen the Institute’s presence across digital and print media, including the development of podcast and broadcast facilities. Played a key role in enhancing the Institute’s branding, digital visibility, media outreach, and overall public perception.
- Serving as **Overall Laboratory Coordinator** of the ECE Department since 2015, coordinating laboratory activities, documentation, maintenance, and inspection requirements in collaboration with Lab In-charges and the HOD.
- Serving as Convener Central purchase Committee (2026-27), MSIT.

Key Publications:

1. **Aman Dahiya** et al., “An Efficient DRL-Dual-Conceptualized Firefly-Weighted Grey Wolf Optimization Approach for Clustering and Routing in IoT Cloud Environmental Wireless Sensor Network,” International Journal of Communication Systems 39, no. 9 (2026). **SCIE**
2. **Dahiya, A., et al.** Design and Construction of a Low Loss Substrate Integrated Waveguide (SIW) for S Band and C Band Applications. *MAPAN* **36**, 355–363 (2021). **SCI-E Indexed**
3. **Dahiya, A., Kumar, D.** A W-band Substrate Integrated Waveguide (SIW) Bandpass Filter at 95 GHz for Millimeter Wave Applications. *Wireless Pers Commun* (2021). **SCI-E Indexed**
4. **Aman Dahiya**, et.al., "Detecting Crop Health using Machine Learning Techniques in Smart Agriculture System" *JSIR*, vol. 80, pp. 699-706, 2021. **SCI-E Indexed**
5. **Dahiya, Aman**, et.al., An Optimized Approach for Feature Extraction in Multi-Relational Statistical Learning, *JSIR*, Vol. 80, No. 6, 2021. **SCI-E Indexed**
6. Chaudhary, K.P., Dubey, P.K., Gahlot, **Dahiya, A.**, Advances in Sensors and Measurements for Metrological Applications. *MAPAN* 36, 211–213 (2021). **SCI-E Indexed**
7. Anand Pratap Singh Sengar, **Aman Dahiya**,”Reconfigurable smart broadband Antenna for wireless Communication Devices, *IEEE Microwave Magazine*, Vol. 17, no. 7, pp. 89-92, July 2016. **SCI-E Indexed**
8. **Kumari, A., Kumar, D. & Kumar, A.** Dual-band substrate integrated waveguide (SIW) band pass filter for scientific radar applications. *Int. j. inf. tecnol.* 11, 875–878 (2019). **Scopus Indexed**

9. Ankur Gupta, Sinjan Kumar, B. Sathya Bama, **Aman Dahiya**, P. Santhosh Kumar, Badugu Samatha,. 2024. "Deep Reinforcement Learning for Dynamic Resource Allocation in IoT-Enabled Big Data Networks". *International Journal of Intelligent Systems and Applications in Engineering* 12 (21s):1136-45. **(Scopus Indexed WOS)**
10. Gupta, A., Anand, R., Sindhwani, N. **Dahiya Aman et al.** Performance and Accuracy Enhancement of Machine Learning & IoT-based Agriculture Precision AI System. *SN COMPUT. SCI.* 5, 930 (2024). **(SCOPUS Indexed)**

Papers presented in Conferences:

1. Aman Dahiya et al., "Predictive Modelling based Cardiovascular Disease Risk Assessment using Machine Learning on Physical Examination Data," 5th International conference on Data Science and Applications (ICDSA 2024), Malaviya National Institute of Technology Jaipur, INDIA, 17th to 19th July, 2024
2. **Aman Dahiya**, Deepti Deshwal, "Design Fundamentals: : Iris Waveguide Filters vs. Substrate Integrated Waveguide (SIW) Bandpass Filters," International conference on Artificial Intelligence and applications (ICAIA-2020), New Delhi, 6-7 Feb 2020.
3. **Aman Dahiya**, Anand Pratap Singh, Dharendra Kumar "Design Techniques of Substrate Integrated Waveguide (SIW) and its Microstrip Transitions," International multidisciplinary conference on current research trends-2020 (**IMCCRT-2020**), 19-20th September, **Malaysia**, 2020.
4. **Aman Dahiya**, Anand Pratap Singh Sengar, Dharendra Kumar, Arun Kumar, "A Critical Review of Substrate Integrated Waveguide for Microwave Applications," International Conference on Computational Intelligence & Communication Technology (CICT-2016) **IEEE computer society**, pp. 495-499 , Ghaziabad, India Feb. 2016.
5. **Aman Kumari**, Adarsh Pal, Dharendra Kumar, "3.5 GHz Microstrip Transmission Line Design for Microwave IC's," International Conference on Latest Trends in Electronics & Communication Engineering, IIMT College of Engineering, Greater NOIDA, 9-10th April 2019.
6. Dharendra Kumar, **Aman Dahiya**, "Compact E-Shaped Microstrip Bandstop Filter," International Centre For Radio Sciences (ICRS), Jodhpur, Rajasthan, 14-17 December 2010.
7. **Aman Kumari**, Arun Kumar, Deepti Deshwal, Neelam ohlan, "A Review on Substrate Integrated Waveguide (SIW) and its Microstrip Transitions," National conference on Advancement in Microwave and Photonic Devices for Electronic communication Systems, RKGIT, Ghaziabad, March 29-31, 2017.
8. **Aman Kumari**, Dharendra Kumar, Arun Kumar, "A Compact E-Shaped Microstrip Bandstop Filter for Wireless Communications," National conference on Advancement in Microwave and Photonic Devices for Electronic communication Systems, RKGIT, Ghaziabad, March 29-

31, 2017.

9. **Aman Kumari**, Manpreet Sandhu, Bhargavi Rohil, Priyanka Dogra, Arun Kumar,” A Low Loss Substrate Integrated Waveguide (SIW) Structure designed for Wireless Communication Systems,” National conference on Advancement in Microwave and Photonic Devices for Electronic communication Systems, RKGIT, Ghaziabad, March 29-31, 2017.
10. **Aman Kumari**, Dharendra Kumar, “E-Shaped Microstrip Bandstop Filter,” Advances in VLSI, Embedded & Communication(AVEC-2010), AFSET, Faridabad, 10th August, 2010.
11. **Aman Kumari**, “Micro Electronics,” National Conference on Information Technology: Setting Trends in Modern Era, N.C. College of Engineering and Technology, Israna, Panipat, March 18-20, 2006.

Patent/Copyright:

1. **Dr. Aman Dahiya**, granted a patent titled “Artificial Intelligence-Based E-Commerce & Business Analytics Kiosk” on 11 December 2025.
2. **Dr. Aman Dahiya**, et.al. has published Patent on “Antenna Development for optimized Radiation” in December 2020.
3. **Dr. Aman Dahiya**, et.al. has granted Patent on “Artificial intelligence based smart detection of Lung disease from chest X-Ray” in March 2021.
4. **Dr. Aman Dahiya**, et.al. has granted Patent on “IOT based smart wearable suit for self health assessment in post COVID era” in July 2021.

Sponsored Project/Consultancy:

1. Completed successfully **IEEE MTT-S** sponsored research project (2014-2019) on “Reconfigurable smart Antenna for wireless Communication Devices.”

Book Chapter/Books published:

2. **Dr. Aman Dahiya**, Designing and Feeding Techniques of Microstrip Patch Antenna, Advance Academic Publisher, 2019. (ISBN no. 978-93-87396-22-7)
3. **Dahiya A.**, Deshwal D. (2021) Design Fundamentals: Iris Waveguide Filters Versus Substrate Integrated Waveguide (SIW) Bandpass Filters. In Advances in Intelligent Systems and Computing, vol 1164. Springer, Singapore.

Memberships of Professional bodies: Life member ISTE, IEEE